```python
from collections import deque

def is_valid(missionaries, cannibals):
    # Number of people on the opposite side
    m_right = 3 - missionaries
    c_right = 3 - cannibals

    # Left side condition
    if missionaries > 0 and cannibals > missionaries:
        return False

    # Right side condition
    if m_right > 0 and c_right > m_right:
        return False

    return True


def missionaries_cannibals():
    # State: (Missionaries, Cannibals, Boat)
    # Boat: 0 = Left, 1 = Right

    start = (3, 3, 0)
    goal = (0, 0, 1)

    queue = deque()
    queue.append((start, []))

    visited = set()

    # Possible boat combinations
    moves = [
        (1, 0),  # 1 Missionary
        (2, 0),  # 2 Missionaries
        (0, 1),  # 1 Cannibal
        (0, 2),  # 2 Cannibals
        (1, 1)   # 1 Missionary and 1 Cannibal
    ]

    while queue:
        state, path = queue.popleft()

        if state in visited:
            continue

        visited.add(state)
        path = path + [state]
```

```
        if state == goal:
            return path

        m, c, boat = state

        for dm, dc in moves:

            if boat == 0:  # Boat moves from left to right
                new_m = m - dm
                new_c = c - dc
                new_boat = 1

            else:  # Boat moves from right to left
                new_m = m + dm
                new_c = c + dc
                new_boat = 0

            if 0 <= new_m <= 3 and 0 <= new_c <= 3:
                if is_valid(new_m, new_c):
                    new_state = (new_m, new_c, new_boat)

                    if new_state not in visited:
                        queue.append((new_state, path))

    return None


solution = missionaries_cannibals()

print("Solution:")

for step, state in enumerate(solution):
    m, c, boat = state

    side = "Left" if boat == 0 else "Right"

    print("Step", step, ":", state, "Boat =", side)
```

IDLE Shell 3.10.0

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```
Python 3.10.0 (tags/v3.10.0:b494f59, Oct  4 2021, 19:00:18) [MSC v.1929 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
=============== RESTART: C:/Users/hi/OneDrive/Desktop/AI/expp.py ===============
Solution:
Step 0 : (3, 3, 0) Boat = Left
Step 1 : (3, 1, 1) Boat = Right
Step 2 : (3, 2, 0) Boat = Left
Step 3 : (3, 0, 1) Boat = Right
Step 4 : (3, 1, 0) Boat = Left
Step 5 : (1, 1, 1) Boat = Right
Step 6 : (2, 2, 0) Boat = Left
Step 7 : (0, 2, 1) Boat = Right
Step 8 : (0, 3, 0) Boat = Left
Step 9 : (0, 1, 1) Boat = Right
Step 10 : (1, 1, 0) Boat = Left
Step 11 : (0, 0, 1) Boat = Right
>>>
```

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expp.py - C:/Users/hi/OneDrive/Desktop/AI/expp.py (3.10.0)

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def missionaries_cannibals():
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    # Boat: 0 = Left, 1 = Right

    start = (3, 3, 0)
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        (0, 2),  # 2 Cannibals
        (1, 1)   # 1 Missionary and 1 Cannibal
    ]

    while queue:
        state, path = queue.popleft()

        if state in visited:
            continue

        visited.add(state)
        path = path + [state]

        if state == goal:
```

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